JOMHACK
EMPOWERING INNOVATION

```
> bootcamp.learn()
```



JomCode Coding Bootcamp

CHAPTER 12

PostgreSQL and SQL fundamentals

```
learn / type / run / explain
```

Plain meaning and technical explanation

PLAIN MEANING

A database stores records in tables so they can be queried and changed.

TECHNICAL EXPLANATION

SQL defines table structures and reads or modifies rows. Primary keys identify rows. WHERE limits affected rows; ORDER BY controls order.

REAL-WORLD USE

Store posts persistently instead of keeping them in a Python list.

QUICK TIP

Run these examples in a practice Supabase project, using SQL Editor.

INSTALL / RUN

Create a free Supabase project, open SQL Editor, paste the complete workbook and Run. Temporary practice tables disappear at transaction end.

Worked example 1: Insert records

workbook.sql

```
01  insert into lesson_posts (id, title, views)
02  values (1, 'Hello', 3), (2, 'Launch', 12);
03  select count(*) as post_count from lesson_posts;
```

PRO TIP SQL text uses single quotes.

Run and explain example 1

EXPECTED RESULT

post_count = 2

WHY IT BEHAVES THIS WAY

Named columns specify how each inserted value maps to the table.

Worked example 2: Select and filter

workbook.sql

```
01    select id, title
02    from lesson_posts
03    where views >= 10
04    order by id;
```

PRO TIP SELECT * is useful while exploring; named columns make contracts clearer.

Run and explain example 2

EXPECTED RESULT

id = 2, title = Launch

WHY IT BEHAVES THIS WAY

WHERE filters rows before the selected columns are returned.

Worked example 3: Update a record

workbook.sql

```
01    update lesson_posts
02    set title = 'Launch day'
03    where id = 2;
04    select title from lesson_posts where id = 2;
```

PRO TIP Check your WHERE clause before running UPDATE or DELETE.

Run and explain example 3

EXPECTED RESULT

title = Launch day

WHY IT BEHAVES THIS WAY

The primary-key condition targets exactly one existing record.

Worked example 4: Delete and inspect

workbook.sql

```
01    delete from lesson_posts where id = 1;
02    select id, title from lesson_posts order by id;
```

PRO TIP An absent matching row is different from a SQL execution error.

Run and explain example 4

EXPECTED RESULT

One row remains: 2, Launch day

WHY IT BEHAVES THIS WAY

DELETE removes rows; the table structure stays available.

Exercise 1: Add a new post

YOUR TASK

Insert id 3 with title Learning and 5 views, then select its title.

CHECK THESE CASES

Exactly one id=3 row exists and views is 5.

HINT AFTER A TRY

Use INSERT followed by SELECT with WHERE.

Exercise 2: Find the most viewed post

YOUR TASK

Return the title of the single most viewed remaining post.

CHECK THESE CASES

Launch day is first with 12 views.

HINT AFTER A TRY

ORDER BY views DESC, then LIMIT 1.

Checkpoint before moving on

01 Run and explain all four worked examples. Match their actual results to the expected behavior.

02 Attempt both exercises before reading the solutions. Verify every acceptance case.

03 Explain how this chapter supports Postlight: Store posts persistently instead of keeping them in a Python list.

Official reference